

AI Video Summarization Project using Mixtral, Whisper, and AWS

Project Overview

Overview

In today's fast-paced digital landscape, there is a significant demand for efficient educational tools. Traditional learning methods are rapidly evolving, with an increasing prevalence of video-based content. However, the abundance of educational videos available online presents a challenge for learners trying to extract key concepts effectively. This highlights the need for a comprehensive tool that can summarize videos and generate quizzes to help learners understand and retain essential information.

The emergence of large language models (LLMs) like Mixtral and Whisper is transforming the landscape of educational technology platforms. These powerful models use advanced natural language processing techniques to quickly understand and process complex educational content.

This project aims to address the growing need for effective educational tools by developing a robust video summarization and quiz generation application powered by state-of-the-art LLMs. By leveraging technologies like Mixtral and Whisper, the platform will extract key insights from educational videos, condense them into concise summaries, and generate quizzes to test comprehension. The project involves implementing a user interface and running the application on AWS cloud infrastructure, including integrating feedback mechanisms to continuously improve the learning experience.

Prerequisite Project: Kindly ensure the completion of the [LLM Project for building and fine-tuning a large language model](#) before proceeding with this project.

Note:

Please note that the use of the OpenAI API may require the utilization of allocated free credits or additional purchases (depending on the account status) for implementing the project. Kindly take note of the free credit limit provided for your usage.

Utilizing AWS services for this project may result in charges; it is essential to thoroughly review the AWS documentation to understand the pricing structure and potential costs associated with different resources and usage patterns.

Aim

The aim of the project is to develop a video summarization and quiz generation tool that leverages Large Language Models and AWS. By integrating advanced LLMs like Mixtral and Whisper, the application will extract key concepts from educational videos, generate concise summaries, and create quizzes to assess comprehension, thereby enhancing the efficiency and effectiveness of learning experiences.

Data Description

The dataset comprises tutorial videos covering diverse topics such as linear regression and other ML concepts.

Tech Stack

- Language: Python 3.10.4
- Libraries: Flask, FFmpeg, Torch, Transformers, SQLAlchemy
- Model: Mixtral, Whisper
- Cloud Platform: Amazon Web Services

Approach

- Initial Setup:
 - Create AWS EC2 instance with GPU for efficient functioning
 - Create environment and Install necessary dependencies and libraries.
- Model Selection:
 - Select Whisper for audio transcription
 - Choose Mixtral for video summarization and quiz generation
- Transcription Functionality:
 - Extract audio from uploaded videos using FFmpeg
 - Transcribe audio to text using the Whisper model
- Quiz Generation:
 - Use Mixtral to summarize and generate quizzes based on transcribed text
 - Implement different complexity levels for quizzes
- Frontend and API Development:
 - Design a user-friendly interface using HTML and JavaScript

- Implement file upload functionality for videos
- Feedback Mechanism:
 - Develop a feedback form for user input
 - Store feedback data in a database for analysis

Cost Breakdown

Components:

- LLM models (Whisper for transcription and Mixtral for summarization and quiz generation).
- Hosted on AWS EC2 g5.12xlarge instances with GPU support.
- Requires data storage (S3) for video files and outputs.

Cost Breakdown as of January 17, 2025

The cost might have been updated, it is recommended to go through the following pricing pages to come up with updated guessestimates.

Open AI Pricing <https://openai.com/api/pricing/>

AWS Pricing <https://aws.amazon.com/ec2/capacityblocks/pricing/>

Component	Description	Usage	Estimated Cost
API Calls (based on model usage)	OpenAI Whisper API for transcription	- Duration: 5-minute video Pricing: \$0.006 per minute	Total: \$0.03 per video
Compute Resources	AWS EC2 g5.12xlarge instance with GPU support	1 instance, runs approx. for 15 minutes per video (for a 5-minute video covering transcription, summarization, and quiz generation).- Pricing: \$5.672/hour	Total: ~\$3.00 per video

Storage	S3 storage for video files and intermediate outputs	Approx. 200 MB per video- Pricing: \$0.023 per GB	Total: \$0.0046 per video
Network Usage	Data transfer costs for S3	Approx. 200 MB per video- Pricing: \$0.09 per GB	Total: \$0.018 per video

Total Costs:

- **Using Open-source Whisper Model:** ~\$1.44 per video (Duration - 5 mins)
- **Using OpenAI Whisper API:** ~\$3.00 per video including storage & network costs

Recommendation: For each 5-minute video, processing with the OpenAI Whisper API will cost approximately \$3.00, covering transcription, video storage, and data transfer. This option is more streamlined and cost-efficient compared to using the open-source Whisper model on AWS EC2, which would cost around \$1.44 per video but requires more management and setup. Also you can choose a smaller instance if you are using whisper API. If ease of use and cost efficiency are a priority, the OpenAI Whisper API is the recommended choice.

Modular code overview:

Once you unzip the code_folder.zip file, you can find the following:

```
├─ app
│   ├── database.py
│   ├── data_models
│   │   ├── feedback.py
│   │   └── __init__.py
│   ├── mixtral_prompts.py
│   ├── mixtral_wrapper.py
│   ├── routes.py
│   ├── templates
│   │   └── index.html
│   ├── utils.py
│   ├── whisper_wrapper.py
│   └── __init__.py
├─ finetune.py
├─ readme.md
├─ requirements.txt
└─ run.py
```

Here is a brief information on the files:

- The code folder contains all the Python scripts and sample files used in the project.
- The README.MD and the solution methodology file provides essential information about the project, usage, and setup instructions.
Kindly follow all the instructions for running the code from Readme.md file
- The requirements.txt file has all the required libraries with respective versions.
Kindly install the file by using the command **pip install -r requirements.txt**

Project Takeaways

1. Understand the importance of video summarization and quiz generation

2. Gain insights into the decision-making process behind selecting language models and tools
3. Understand the strengths and limitations of popular language models like GPT-3, BERT, and Mixtral
4. Understand the impact of GPU-equipped instances on reducing latency
5. Learn how to set up AWS EC2 instances for running large language models effectively
6. Learn to integrate large language models like Mixtral and Whisper
7. Gain insights into the process of extracting audio from videos using FFmpeg
8. Understand the steps involved in transcribing audio to text using the Whisper model
9. Learn to generate quizzes dynamically based on various prompting strategies
10. Explore the implementation of API and front-end development using Flask, HTML, and JavaScript.
11. Gain experience in storing user feedback data in a database for feedback mechanism